

Day 5 - Decision Tree Classification

Temperature	Vibration	Failure
High	High	Yes
High	Low	Yes
Medium	High	Yes
Low	Low	No
Low	Medium	No
Medium	Low	No
High	Medium	Yes
Low	High	No

Input Data:

* Temperature = High
* Vibration : Medium

Total = 8

Failure = 4 = (P_1)

Non failure = 4 = (P_2)

$P_{yes} = P_{no} = 0.5$

$$\text{Entropy}(S) = - \sum_{i=1}^n P_i \log_2 P_i$$

$$= -P_1 \log_2 P_1 - P_2 \log_2 P_2 \Rightarrow P_{yes} \log_2 P_{yes} - P_{no} \log_2 P_{no}$$

$$= -0.5 \log_2 (0.5) - (0.5) \log_2 (0.5)$$

$$= 1$$

$$\text{Entropy} = 1$$

Information Gain for temperature

Temperature = High:

$$\text{Failure (Yes)} = 3$$

$$\text{Non Failure (No)} = 0$$

$$\text{Entropy (High)} = 0$$

Temperature = Medium:

$$\text{Failure (Yes)} = 1$$

$$\text{Non Failure (No)} = 1$$

$$\text{Entropy (Medium)} = 1$$

Temperature = Low:

$$\text{Failure (Yes)} = 0$$

$$\text{Non Failure (No)} = 3$$

$$\text{Entropy (Low)} = 0$$

$$\text{Weighted Entropy (Temp)} = \frac{3}{8} \times 0 + \frac{2}{8} \times 1 + \frac{3}{8} \times 0$$

$$\text{WE (Temp)} = \underline{\underline{0.25}}$$

$$\text{Information Gain (Temp)} = \text{Entropy} - \text{WE}$$

$$= 1 - 0.25$$

$$= \underline{\underline{0.75}}$$

Information Gain For Vibration

Vibration = Low:

Failure (Yes) = 1

Non Failure (no) = 1

Entropy (Low) = 1

Vibration = Medium:

Failure (Yes) = 1

Non Failure (no) = 1

Entropy (medium) = 1

Vibration = High:

Failure (Yes) = 2

Non Failure (no) = 1

Entropy = $-\frac{2}{3} \log_2 (\frac{2}{3}) - \frac{1}{3} \log_2 (\frac{1}{3})$

Entropy (high) ≈ 0.918

Weighted Entropy (Vib) = $\frac{3}{8} \times 0.918 + \frac{2}{8} + \frac{2}{8}$

= 0.844

Information Gain (Vib) = $1 - 0.844$

= 0.156

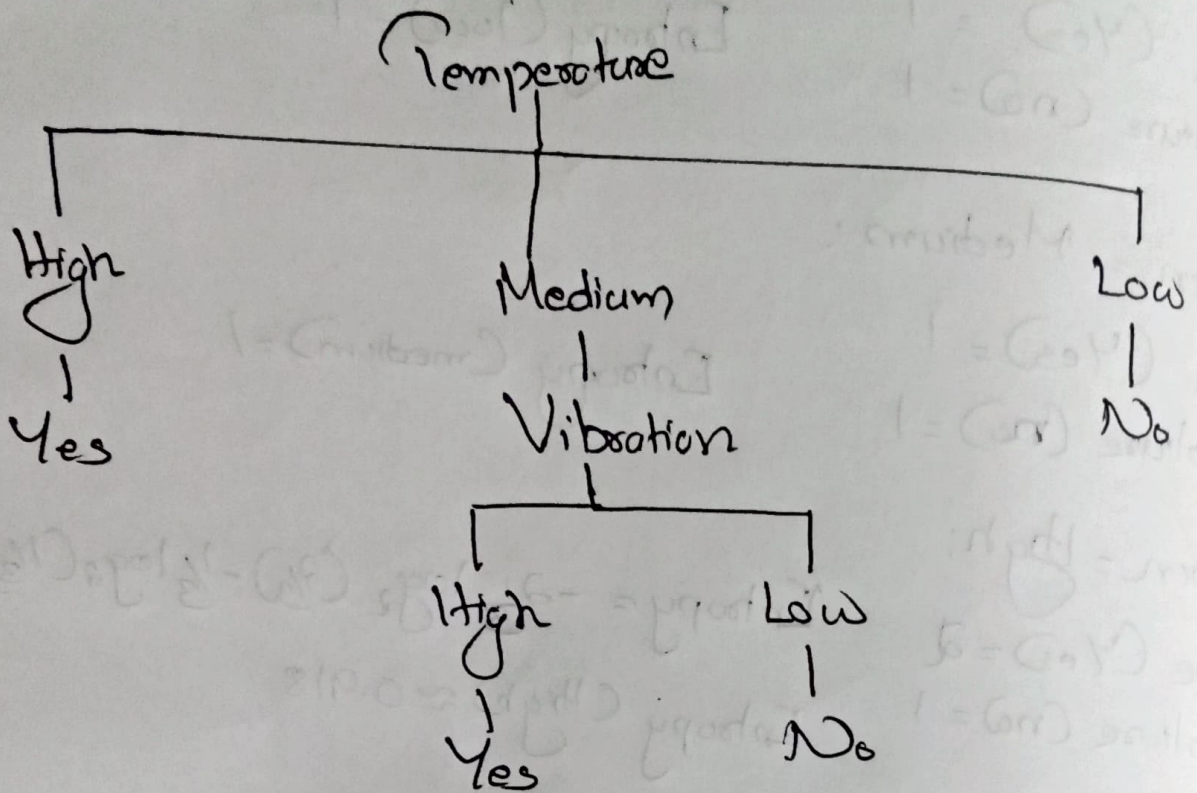
Choosing Root Node

Temperature = 0.75

Vibration = 0.156

$IG(Temp) > IG(Vib)$

Tree



Prediction:

Input value $\Rightarrow$ Temperature = High, Vibration = Medium
Failure =

Temperature = High $\rightarrow$ Yes [Failure]

Result

Machine will be fail.

FileEditSelectionViewGoRunTerminalHelp

AL_ML_INTERNSHIP

EXPLORER

AL_ML_INTERNSHIP

- Day1
- Day2
- Day3
 - code_3.py
 - Fuel_Consumption_model.pkl
 - KNN_Dataset.csv
 - Report_Day3.pdf
- Day4
- Day4.2
- Day5
 - machine_failure_Dtree_model.pkl
- README.md

code_5.py

Decision_Tree.csv

machine_failure_Dtree_model.pkl

Day5

code_5.py

print measure encoding: /

40 for i, label in enumerate(encoder_fail.classes_):

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS C:\Users\Aleni\OneDrive\Documents\AI_ML_INTERNSHIP> C:\Users\Aleni\AppData\Local\Programs\Python\Python312\python.exe c:/Users/Aleni/OneDrive/Documents/AI_ML_INTERNSHIP/Da

y5/code_5.py

Encoded Dataset:

	Machine	Temperature	Vibration	Failure
0	1	0	0	1
1	2	0	1	1
2	3	2	0	1
3	4	1	1	0
4	5	1	2	0
5	6	2	1	0
6	7	0	2	1
7	8	1	0	0

Temperature Encoding:

High = 0

Low = 1

Medium = 2

Vibration Encoding:

High = 0

Low = 1

Medium = 2

Failure Encoding:

No = 0

Yes = 1

Prediction for New Machine:

Temperature = High

Vibration = Medium

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DECISION TREE RESULT

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Predicted Result: Failure of machine will occur

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main

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Spaces: 4

UTF-8

CRLF

Python

Python 3.12.664-bit

Decision Tree for Machine Failure Prediction

